

The pair singular series of a polynomial V : the dilated pieces below $H^{1/2}$

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Abstract

[Draft. The proofs of Theorems 3 and 5 are in [research/explore/PAPER-V-thmM.md](#) and [PAPER-V-smallend-draft.md](#) and are under adversarial reading; nothing here is to be circulated before those readings are recorded.] For a monic irreducible quadratic f without fixed prime divisor, part III decomposed the Cesàro off-diagonal $\text{Off}_f^*(H)$ into pieces indexed by the dilation u and proved that the pieces with $u > H^{1/2+\varepsilon}$ contribute a main term $c_{\text{off}}(f)H$. This paper treats dilations below that threshold. We prove that the pieces with $H^{0.47} < u \leq H^{0.53}$ contribute $O(H^{1-\delta})$, so that $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta})$; the input is a bilinear estimate for Kloosterman fractions of Bourgain and Garaev together with a bound for the additive energy of reciprocals, valid for every modulus, which we deduce from an argument of Cilleruelo and Garaev. For negative discriminants we prove a second result at the other end: the pieces with $u \leq H^{1/3-c}$ contribute $O(H^{1-\delta(c)})$, by running the automorphic kernel method of Grimmelt and Merikoski with the leading coefficient u^2 tracked. What remains is a neighbourhood of $u = H^{1/3}$.

1 Introduction

[Working draft: the proofs are in [research/explore/PAPER-V-thmM.md](#) and [PAPER-V-smallend-draft.md](#) and are under adversarial reading.]

The problem. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be monic, irreducible, without fixed prime divisor, of discriminant D . Part III [6] wrote the Cesàro off-diagonal of the pair singular series as a sum of *pieces*,

$$\text{Off}_f^*(H) = P(1) \sum_u \lambda(u) \omega(u) \mathcal{P}_u(H/u),$$

one for each squarefree *dilation* u , the piece $\mathcal{P}_u(Y)$ being the same problem at length $Y = H/u$ with the roots of $x^2 \equiv D$ replaced by those of $u^2 x^2 \equiv D$. Each piece is $O(Y)$, and the weights satisfy $\sum_{u \leq H} |w(u)|/u \asymp \log H$: the trivial bound for the whole sum is $H \log H$, which is exactly the size of the statement to be proved. Every piece must therefore be small, and the difficulty is uniformity in u .

Exponent coordinates. Write $u = H^\alpha$. Each method available for the pieces gives a saving $H^{-s(\alpha)}$ over the trivial bound on an interval of α , and degenerates at the ends of that interval. In these coordinates part III proves $s > 0$ for $\alpha > 2/3$ (averaging the dilations along u), and the present author's extension of that theorem, recorded in the revised part III, gives $s > 0$ for $\alpha > 1/2$ by a dispersion argument with Weil's bound. Below $1/2$ both mechanisms fail: the

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incomplete Kloosterman sums that dispersion produces have modulus $[d_1, d_2] \asymp Y^2$, and Weil's bound is trivial once the range of u is shorter than the square root of that modulus. This paper supplies two further mechanisms, one on each side of $\alpha = 1/3$.

Results. Theorem 3 treats α near $1/2$: the weight w is decomposed as $1 * \chi_D * \gamma$, and the resulting bilinear forms in u are estimated by Weil's bound when one factor is long (Type I) or short (Type II), and, in the remaining balanced case, by the bilinear bound of Bourgain and Garaev [1] for Kloosterman fractions. That bound needs an estimate for the additive energy of reciprocals; the one in [1] is too weak at the sizes that occur here, and Lemma 1 supplies a stronger one, valid for every modulus, by transferring an argument of Cilleruelo and Garaev [2] for points on modular hyperbolas. Theorem 5 treats small α for negative discriminants: piece u is, after removing the content, a Type I sum for the polynomial $u^2\ell^2 + |D|$ in the sense of Grimmelt and Merikoski [3], and their automorphic kernel theorem [4] applies at level u^2e once the two kernel quantities are estimated with the dilation tracked. The diagonal is the number of orbits of the family, $\asymp u$; the off-diagonal, which their general bound would give as $\asymp u^2$, is in fact bounded independently of u (Lemma 4), because the distances inside the family are quantised in steps of $1/(4h)$ whatever the dilation.

What remains. Both mechanisms degenerate at $\alpha = 1/3$, from opposite sides and for unrelated reasons: there the balanced Type II atoms sit at $d^{1/4}$, where the energy input is not strong enough, and the kernel method's diagonal $\asymp u$ meets the trivial bound. The final section explains what the residual band costs. Two points are worth recording. A band of width 2δ in α costs $\delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$, because part III's Theorem E applies piece by piece inside it; so a failing method never makes the total worse. And if the savings on both sides vanish only at $\alpha = 1/3$, with implied constants that are powers of $\log H$, then the blocks near the point contribute $O(H)$ in total, which would give $\text{Off}_f^*(H) = O(H)$ — Hypothesis (E) of part I [5], a bounded second-order term in the Cesàro asymptotic. The constant A_f needs strictly more: a saving at the point itself.

2 Notation

We follow part III [6] throughout. [to import: $\lambda(p) = p/(p - 2\omega(p))$, $P(1)$, $w(u) = P(1)\lambda(u)\omega(u)$ supported on squarefree u ; admissible moduli d' (squarefree, split, coprime to $2Du$); $R_{d'}^{(u)}$, ρ_k , $B_{d'}^{(Y)}$, $\mathcal{P}_u$, $G_{d'}(a, Y)$, $\bar{G}_{d'}$, $G^\circ = G - \bar{G}$, the finite Fourier expansion and its coefficients $\xi_k(d', N)$, Lemma kloost, Lemma tail, Theorem small.]

3 The energy of reciprocals and bilinear Kloosterman fractions

Lemma 1 (energy of reciprocals). [PAPER-V-thmM.md Lemma E; proof written, two readings.]

Lemma 2 (Bourgain–Garaev). [cite [1]; Corollary B'.]

4 The pieces around $u = H^{1/2}$

Theorem 3. [Theorem M; proof = Lemmas K, E, B', S and Steps R1–R5.]

5 The small dilations

Lemma 4 (the off-diagonal at one level does not grow with the dilation). [*Lemma A'' of PAPER-V-smallend-draft.md.*]

Theorem 5. [*Theorem S ($D < 0$); proof = S1–S8 with Lemma 4, S1', S2', O5, O6'.]*]

6 What remains

[the band at $u = H^{1/3}$: the two mechanisms degenerate there; the accounting of PROOFS §31–§34; what a positive saving at the point would give (A_f), and what a point gap with polylog losses would give (part I's Hypothesis (E)).]

References

- [1] Jean Bourgain and Moubariz Z. Garaev. Kloosterman sums in residue rings. *Acta Arith.*, 164(1):43–64, 2014.
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